Penetration Testing Curriculum Development in Practice[1]

Abstract

As both the frequency and the severity of network breaches have increased in recent years, it is essential that cybersecurity be incorporated into the core of business operations. Evidence from the U.S. Bureau of Labor Statistics (Bureau of Labor Statistics, 2012) indicates that there is, and will continue to be, a severe shortage of cybersecurity professionals nationwide throughout the next decade. To fill this job shortage we need a workforce with strong hands-on experience in the latest technologies and software tools to catch up with the rapid evolution of network technologies. It is vital that the IT professionals possess up-to-date technical skills and think and act one step ahead of the cyber criminals who are constantly probing and exploring system vulnerabilities. There is no perfect security mechanism that can defeat all the cyber-attacks; the traditional defensive security mechanism will eventually fail to the pervasive zero-day attacks. However, there are steps to follow to reduce an organization's vulnerability to cyber-attacks and to mitigate damages. Active security tests of the network from a cyber-criminal's perspective can identify system vulnerabilities that may lead to future breaches. "If you know yourself but not the enemy, for every victory gained you will also suffer a defeat. But if you know the enemy and know yourself, you need not fear the result of hundred battles" (Sun, 2013). Penetration testing is a discipline within cybersecurity that focuses on identifying and exploiting the vulnerabilities of a network, eventually obtaining access to the critical business information. The pentesters, the security professionals who perform penetration testing, or ethical hackers, break the triad of information security - Confidentiality, Integrity, and Accountability (CIA) - as if they were a cyber-criminal. The purpose of ethical hacking or penetration testing is to know what the "enemy" can do and then generate a report for the management team to aid in strengthening the system, never to cause any real damages. This paper introduces the development of a penetration testing curriculum as a core class in an undergraduate cybersecurity track in Information Technology. The teaching modules are developed based on the professional penetration testing life cycle. The concepts taught in the class are enforced by hands-on lab exercises. This paper also shares the resources that are available to institutions looking for teaching materials and grant opportunities to support efforts when creating a similar curriculum in cybersecurity.

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